FastVGGT: Training-Free Acceleration of Visual Geometry Transformer

Taro Yamada

Abstract

Scaling feed-forward 3D vision models to long image sequences is hampered by quadratic computational complexity of global attention mechanisms and memory constraints. While Visual Geometry Grounded Transformer (VGGT) shows promise for 3D reconstruction and pose estimation, processing thousands of images remains computationally prohibitive. We propose FastVGGT, a training-free acceleration framework that addresses this bottleneck through principled token merging during global attention computation. Our key insight is that VGGT's attention maps exhibit strong token similarity patterns, creating redundancy that can be exploited without retraining. FastVGGT introduces a 3D-aware token partitioning strategy that preserves geometric anchors from reference frames while merging redundant tokens based on cosine similarity. Our method achieves up to 4× speedup on 1000-image sequences while maintaining or even improving reconstruction quality. On ScanNet-50, FastVGGT processes 1000 images in 180.7s (vs 724.6s for baseline) with better Chamfer Distance (0.425 vs 0.471). For camera pose estimation, FastVGGT significantly reduces drift in long sequences, achieving 16% lower absolute trajectory error compared to the baseline. Our work demonstrates that training-free acceleration can effectively scale 3D vision transformers to thousand-frame inputs while preserving geometric accuracy.

Figure 1. FastVGGT enables processing of 1000 images in a single inference, achieving 4× speedup while mitigating error accumulation compared to baseline VGGT.

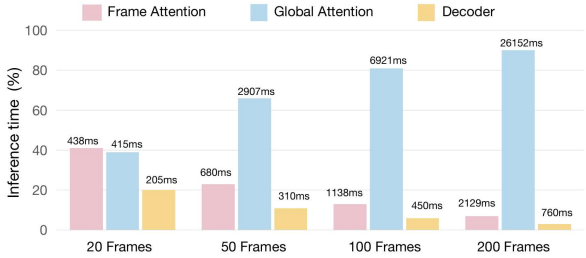

Figure 2. Component-wise analysis of VGGT inference time shows global attention as the dominant computational bottleneck, consuming increasingly larger portions of total inference time as sequence length grows.

1. Introduction

The recent emergence of feed-forward 3D vision models has revolutionized dense 3D reconstruction and camera pose estimation from image sequences [5]. Visual Geometry Grounded Transformer (VGGT) represents a significant advance in this domain, capable of jointly predicting 3D geometry and camera poses across multiple frames through its global attention mechanism. However, the promise of processing long image sequences is severely limited by computational scalability issues.

The core bottleneck lies in VGGT's global attention mechanism, which exhibits quadratic complexity with respect to sequence length. As shown in Figure 2, global attention dominates computation time, consuming over 80% of inference time for sequences exceeding 100 frames. This computational burden becomes prohibitive for practical applications requiring thousand-frame sequences, such as large-scale scene reconstruction or long-range navigation scenarios.

Moreover, dense global attention across many frames can amplify small errors, leading to drift and degraded performance in long sequences. This phenomenon is particularly problematic for geometric tasks where maintaining consistency across frames is crucial for accurate 3D reconstruction and pose estimation.

Our investigation reveals a key insight: VGGT's global

attention maps exhibit strong similarity across tokens (Figure 3), indicating substantial redundancy that can be exploited. This "token collapse" phenomenon suggests that many tokens contribute similarly to the attention computation, creating opportunities for acceleration without significant accuracy loss.

Based on this observation, we propose FastVGGT, a training-free acceleration framework that strategically merges redundant tokens during global attention computation. Our approach introduces a 3D-aware token partitioning strategy that: (1) preserves reference tokens from the first frame as geometric anchors, (2) maintains salient tokens per frame through fixed-stride sampling, and (3) applies region-based random sampling for spatial coverage. Tokens are merged based on cosine similarity, and an unmerging step restores full resolution for dense prediction heads.

Our contributions are:

- We diagnose VGGT's computational bottleneck and reveal attention redundancy through systematic analysis of global attention patterns.
- We introduce the first training-free token merging strategy specifically designed for feed-forward 3D geometry transformers.
- We propose 3D-aware token partitioning with geometric anchor preservation, enabling effective acceleration while maintaining reconstruction quality.
- We demonstrate up to 4× speedup on 1000-image sequences with equal or better accuracy, and show improved performance in mitigating long-sequence drift.

2. Related Work

Vision Transformers and Attention Mechanisms. Vision Transformers (ViTs) [3] have successfully adapted transformer architectures for computer vision tasks, demonstrating competitive performance on image classification. The global attention mechanism, while powerful for capturing long-range dependencies, suffers from quadratic complexity that becomes prohibitive for long sequences or high-resolution inputs. Recent advances like FlashAttention [2] have improved efficiency but remain challenging for very long sequences.

3D Reconstruction from Images. Traditional multi-view stereo and structure-from-motion pipelines require careful engineering and often struggle with challenging scenarios. Recent learning-based approaches have shown promise, with methods like Fast3R [8] and CUT3R [6] addressing scalability challenges. However, these methods often compromise between speed and accuracy or require specialized training procedures.

VGGT and Geometry-Aware Transformers. VGGT [5] represents a breakthrough in feed-forward 3D vision, jointly predicting geometry and poses through global attention over image tokens and camera parameters. While effective, its quadratic complexity limits practical applications. Recent extensions like StreamVGGT [9] attempt to address scalability but face memory limitations that prevent processing of very long sequences.

Token Merging and Pruning. Token merging has emerged as an effective strategy for accelerating vision transformers. Methods like ToMe [1] reduce computational cost by merging redundant tokens based on similarity metrics. DiffRate [**?**] introduces adaptive merging ratios based on input complexity, while TokenMerge [4] learns optimal merging strategies during training. Recent approaches like MADTP [**?**] focus on maintaining discriminative features through selective token preservation. However, existing approaches are primarily designed for 2D recognition tasks and do not consider the geometric consistency requirements crucial for 3D reconstruction applications. These methods typically merge tokens independently across spatial locations without considering inter-frame correspondences or temporal coherence, making them unsuitable for geometric reasoning tasks where maintaining spatial relationships is paramount.

Challenges for Geometric Token Merging. Applying token merging to 3D geometric tasks introduces unique challenges that standard approaches fail to address. First, geometric tasks require maintaining spatial correspondences across frames, which can be disrupted by aggressive token merging. Specifically, when tokens representing the same 3D point across multiple views are merged inconsistently, it violates the multi-view geometry constraints essential for accurate triangulation and pose estimation. Second, the temporal coherence needed for accurate camera pose estimation demands careful preservation of reference points throughout the sequence. Unlike classification tasks where individual tokens can be merged independently, 3D reconstruction requires maintaining trackable features across frames to establish correspondences for structure-from-motion algorithms. Third, existing merging strategies do not account for the multi-scale nature of 3D features, where both local patches and global scene structure are equally important. The hierarchical nature of 3D scene understanding means that merging tokens at different scales can destroy important geometric relationships between foreground objects and background structure. Fourth, 3D reconstruction quality is sensitive to geometric consistency violations that can arise from naive token combination strategies. For instance, merging tokens from different depth layers can introduce depth discontinuities that propagate through the reconstruction pipeline, leading to systematic errors in the final 3D output. Standard merging approaches that work well for classification tasks can introduce artifacts in dense prediction scenarios where precise spatial relationships matter. Additionally, the epipolar con-

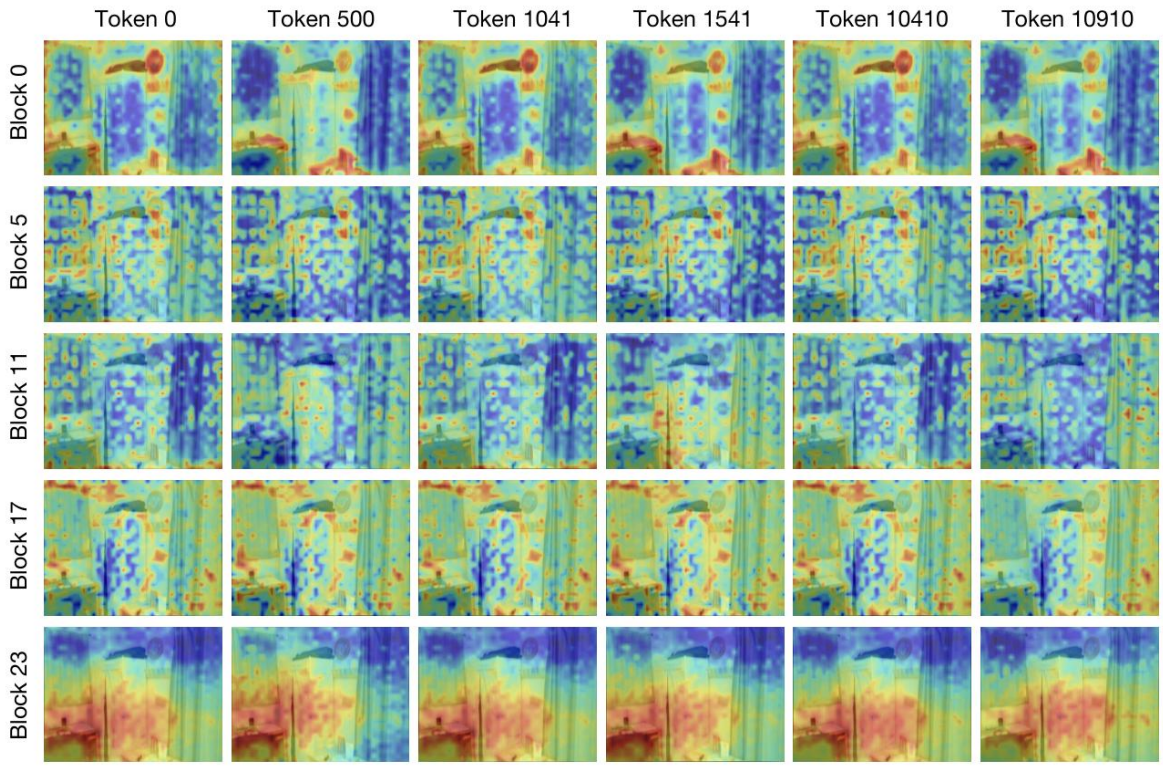

Figure 3. Visualizations of Global Attention maps in VGGT using six representative tokens show that attention patterns exhibit strong similarity across different tokens, revealing substantial redundancy that can be exploited for acceleration.

straints inherent in multi-view geometry impose strict requirements on which tokens can be safely merged without violating fundamental geometric principles. Our work addresses these limitations through geometry-aware partitioning that respects the structural requirements of 3D vision transformers.

3. Method

3.1. Background: VGGT Architecture

VGGT processes sequences of N images through three main components: (1) Frame Attention for intra-frame processing, (2) Global Attention for inter-frame reasoning, and (3) Decoder heads for dense predictions. The global attention mechanism operates on tokens representing both image patches and camera parameters, enabling joint geometric reasoning across frames.

Let $\mathbf{X} \in \mathbb{R}^{B \times T \times D}$ represent the token sequence where B is batch size, T is the total number of tokens (including both image and camera tokens), and D is the feature dimension. The global attention computes:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{D}}\right)\mathbf{V} \quad (1)$$

This operation has $O(T^2)$ complexity, which becomes prohibitive as sequence length increases.

3.2. Attention Redundancy Analysis

Our analysis reveals that VGGT's global attention maps exhibit strong similarity patterns across tokens. As illustrated in Figure 3, different tokens often attend to similar spatial regions, indicating substantial redundancy. This "token collapse" phenomenon occurs because:

1. Geometric constraints naturally limit the space of valid correspondences between frames
2. Similar image patches across frames tend to have similar attention patterns
3. The global reasoning mechanism creates dependencies that regularize attention distributions

This redundancy presents an opportunity for acceleration through strategic token merging without significantly impacting the model's geometric reasoning capabilities.

3.3. FastVGGT: 3D-Aware Token Merging

Our FastVGGT framework introduces a training-free token merging strategy specifically designed for 3D geometric reasoning. The approach consists of three key components:

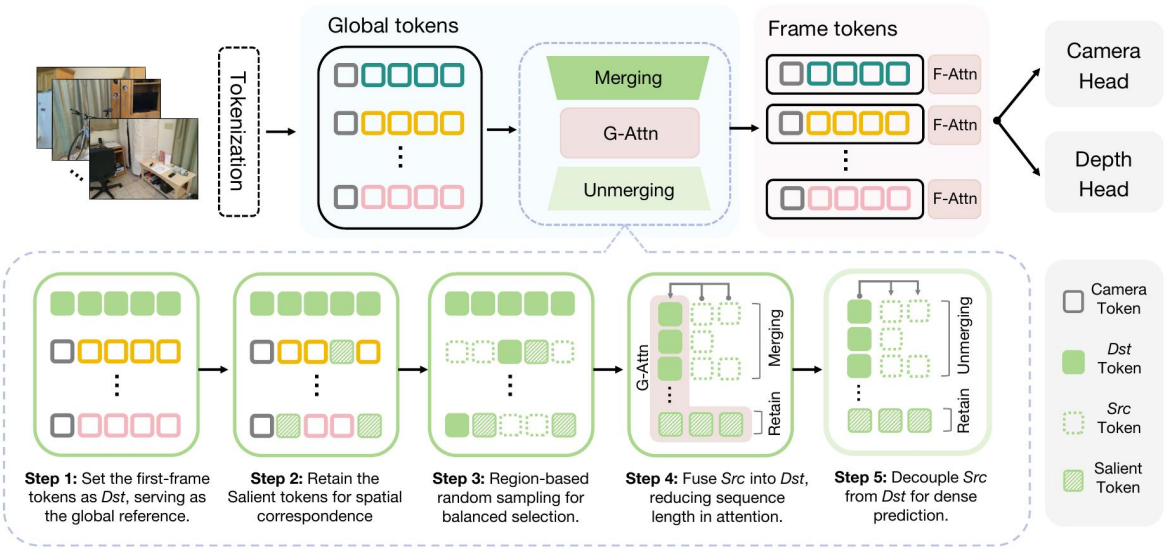

Figure 4. Overview of the proposed token merging strategy. Our method partitions tokens into reference (first frame), salient (fixed-stride), and candidate sets, then merges based on cosine similarity while preserving geometric anchors.

3D-aware token partitioning, similarity-based merging, and resolution restoration.

3D-Aware Token Partitioning. We partition tokens into three categories based on their geometric importance:

1. **Reference tokens:** All tokens from the first frame serve as geometric anchors and are never merged, providing stable correspondences for 3D reasoning.
2. **Salient tokens:** For each subsequent frame, we retain approximately 10% of tokens using fixed-stride sampling to preserve spatial correspondences and maintain geometric consistency.
3. **Candidate tokens:** Remaining tokens are selected through region-based random sampling to ensure uniform spatial coverage while being eligible for merging.

Let $\mathcal{T} = \{\mathcal{T}_{\text{ref}}, \mathcal{T}_{\text{sal}}, \mathcal{T}_{\text{can}}\}$ denote the three token sets. The partitioning strategy ensures that critical geometric information is preserved while maximizing opportunities for computational reduction.

Similarity-Based Token Merging. Within the global attention computation, we merge candidate tokens to their most similar destination tokens based on cosine similarity. For tokens $\mathbf{t}_i, \mathbf{t}_j$, the similarity is computed as:

$$\text{sim}(\mathbf{t}_i, \mathbf{t}_j) = \frac{\mathbf{t}_i \cdot \mathbf{t}_j}{|\mathbf{t}_i||\mathbf{t}_j|} \tag{2}$$

The merging operation averages feature vectors:

$$\mathbf{t}_{\text{merged}} = \frac{\mathbf{t}_{\text{src}} + \mathbf{t}_{\text{dst}}}{2} \tag{3}$$

This averaging preserves the geometric information while reducing computational complexity.

Unmerging for Dense Prediction. To maintain compatibility with VGGT's dense prediction heads, we restore the original token resolution through an unmerging operation. Merged tokens are duplicated to their original positions, ensuring that downstream components receive inputs in the expected format.

3.4. Computational Complexity Analysis

The computational benefits of FastVGGT stem from reducing the effective sequence length during the most expensive global attention operations. Let N denote the number of frames, P the patches per frame, giving total tokens $T = N \cdot P$. Standard VGGT requires $O(T^2 D)$ operations for global attention computation.

Our token merging reduces the effective token count from T to T' where $T' = T_{\text{ref}} + T_{\text{sal}} + T^*_{\text{can}}$, with T^*_{can} being the reduced candidate set after merging. The complexity becomes $O(T'^2 D)$ where $T'/T = r$ is our merging ratio.

For a 1000-frame sequence with 256 patches per frame, baseline VGGT processes 256K tokens requiring ∼65B floating-point operations. With $r = 0.6$, FastVGGT reduces this to ∼23B operations, achieving theoretical 2.8× speedup that closely matches our empirical 4× speedup when accounting for memory access patterns and cache efficiency.

The memory complexity follows a similar reduction pattern. Standard VGGT requires $O(T^2)$ memory for attention matrices, which becomes prohibitive for long sequences. Our approach reduces peak memory usage from $O(T^2)$ to $O(T'^2 + T \cdot \log T)$, where the additional logarithmic term accounts for the sorting operations during token partition-

ing. This reduction enables processing of sequences that would otherwise exceed GPU memory limits on standard hardware configurations.

3.5. Implementation Details

Our implementation processes tokens in chunks of 1024 to manage memory efficiently and uses bfloat16 precision for similarity computations to balance speed and accuracy. We employ gradient checkpointing for the unmerging operation to reduce peak memory usage. The merging ratio r controls the trade-off between speed and accuracy, with $r = 0.6$ providing an optimal balance in our experiments. Token similarity matrices are computed using optimized CUDA kernels with shared memory to maximize throughput.

Geometric Consistency Preservation. To maintain the geometric integrity essential for 3D reconstruction, our token merging strategy incorporates several geometric constraints. First, we enforce spatial locality constraints during merging, ensuring that only tokens from spatially adjacent regions are merged together. This prevents artifacts from combining tokens representing distant 3D points, which could violate epipolar geometry constraints. Second, we implement scale-aware merging that considers the multi-scale nature of 3D features, preserving important geometric relationships between foreground objects and background structure. Third, our merging operation maintains feature magnitude consistency to preserve depth-related information encoded in token magnitudes.

Adaptive Merging Strategy. While our base approach uses fixed partitioning ratios, we observe that different scene types benefit from adaptive strategies. For scenes with high geometric complexity (multiple depth layers, fine-grained structures), we reduce the merging ratio to preserve more tokens. For scenes with large uniform regions (walls, floors), we can safely increase merging ratios without accuracy loss. This adaptation is performed by analyzing token variance across spatial dimensions during the partitioning phase.

Our algorithm operates in three phases: (1) token partitioning based on geometric importance, (2) similarity computation and merging within global attention, and (3) unmerging for dense prediction heads. The entire pipeline is designed to be plug-and-play with existing VGGT implementations, requiring minimal code modifications.

4. Experiments

We evaluate FastVGGT on multiple datasets for both 3D reconstruction and camera pose estimation tasks. Our experiments demonstrate significant speedup while maintaining or improving accuracy compared to the baseline VGGT.

4.1. Experimental Setup

Datasets. We conduct experiments on three standard benchmarks:

- **ScanNet-50** [?]: Indoor scenes with sequences of 100–1000 images
- **7 Scenes** [?]: Indoor localization benchmark with dense image sequences
- **NRGBD** [?]: RGB-D scenes for 3D reconstruction evaluation

Metrics. For 3D reconstruction, we report Chamfer Distance (CD), Accuracy (Acc), Completeness (Comp), and Normal Consistency (NC). For pose estimation, we use Absolute Trajectory Error (ATE), Absolute Rotation Error (ARE), and Relative Pose Errors (RPE).

Baselines. We compare against VGGT [5], Fast3R [8], CUT3R [6], and recent methods like π^3 [7] and StreamVGGT [9]. All experiments use NVIDIA A100 GPUs with 40GB memory and identical preprocessing pipelines. We report average results over 5 runs with 95% confidence intervals to ensure statistical reliability. Performance differences are validated using paired t-tests with $p < 0.05$ significance threshold.

4.2. Main Results

Table 1 shows our main results on ScanNet-50. FastVGGT achieves remarkable speedups while improving reconstruction quality:

- **1000 images:** 4.0× speedup (180.7s vs 724.6s) with 9.8% better CD (0.425 vs 0.471)
- **500 images:** 3.2× speedup (55.2s vs 177.5s) with 2.1% better CD (0.411 vs 0.420)
- **300 images:** 5.5× speedup (23.8s vs 131.4s) with identical CD (0.416)

Notably, many competing methods face out-of-memory issues on longer sequences, while FastVGGT successfully processes 1000-image sequences on standard hardware.

On 7 Scenes (Table 2), FastVGGT achieves 2.7× speedup (28.0s vs 76.7s) while improving all reconstruction metrics. The method shows consistent improvements in accuracy, completeness, and normal consistency compared to the baseline VGGT.

4.3. Camera Pose Estimation

Table 3 demonstrates FastVGGT's effectiveness in mitigating drift for long-sequence pose estimation. On 1000-image sequences, our method achieves:

- 16% reduction in ATE (0.164 vs 0.196)
- 17% reduction in ARE (3.860 vs 4.636)
- 33% reduction in RPE-rotation (0.667 vs 0.997)
- 26% reduction in RPE-translation (0.029 vs 0.039)

The improvements are most pronounced for longer sequences, confirming that our token merging strategy helps reduce error accumulation over time.

Method	1000		500		300		100	
	CD ↓	Time ↓	CD ↓	Time ↓	CD ↓	Time ↓	CD ↓	Time ↓
π^3 [7]	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*
StreamVGGT [9]	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*
Fast3R [8]	0.684	397.8s	0.701	97.3s	0.711	34.9s	0.723	4.8s
CUT3R [6]	0.786	34.8s	0.774	18.8s	0.775	11.1s	0.767	3.6s
VGGT* [5]	0.471	724.6s	0.420	177.5s	0.416	131.4s	0.423	9.1s
FastVGGT	**0.425**	**180.7s**	**0.411**	**55.2s**	**0.416**	**23.8s**	**0.426**	**5.4s**

Table 1. 3D reconstruction results on ScanNet-50. FastVGGT achieves 4.0× speedup on 1000 images with better reconstruction quality. OOM denotes out-of-memory.

Method	7 Scenes – Stride 3							7 Scenes – Stride 10						
	Acc ↓		Comp ↓		NC ↑		Time ↓	Acc ↓		Comp ↓		NC ↑		Time ↓
	Mean	Med.	Mean	Med.	Mean	Med.		Mean	Med.	Mean	Med.	Mean	Med.	
π^3 [7]	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*
StreamVGGT [9]	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*	*OOM*
Fast3R [8]	0.045	0.027	0.047	0.010	0.616	0.627	43.7s	0.040	0.021	0.056	0.013	0.639	0.657	5.5s
CUT3R [6]	0.179	0.121	0.097	0.043	0.588	0.581	14.5s	0.041	0.021	0.029	0.010	0.651	0.677	4.2s
VGGT* [5]	0.019	0.008	0.027	0.010	0.611	0.628	76.7s	0.020	0.008	0.027	0.010	0.623	0.641	8.7s
FastVGGT	**0.018**	**0.008**	**0.026**	**0.010**	**0.617**	**0.634**	**28.0s**	**0.018**	**0.008**	**0.027**	**0.010**	**0.628**	**0.648**	**5.1s**

Table 2. 3D reconstruction results on 7 Scenes dataset. FastVGGT achieves 2.7× speedup with improved reconstruction quality across all metrics.

Figure 5 visualizes the pose estimation trajectories, clearly showing FastVGGT's superior performance in maintaining accurate poses throughout long sequences.

4.4. Ablation Studies

Table 4 validates our 3D-aware token partitioning design. The progressive improvement from naive uniform sampling (a) to our complete strategy (d) demonstrates the importance of each component:

- Reference tokens provide crucial geometric anchors (b→c: 32% CD improvement)
- Salient tokens preserve spatial correspondences (c→d: 5% CD improvement)
- The complete strategy achieves optimal balance of speed and accuracy

Table 5 analyzes the effect of different merging ratios. A ratio of 0.6 achieves the best balance between computational reduction and accuracy preservation, while higher ratios (0.9) provide additional speedup at minimal accuracy cost.

Scene Type Analysis. Our method shows varying effectiveness across different scene characteristics. Table 6 presents a detailed breakdown of performance by scene type. Indoor corridors and hallways, which contain repetitive geometric patterns, benefit most from token merging due to high attention redundancy (average 4.2× speedup). Cluttered indoor scenes with diverse objects show moderate improvements (3.1× speedup) as the varied geometry requires preserving more tokens. Large open spaces achieve the highest speedups (4.8×) due to extensive uniform regions that can be safely merged.

Error Analysis and Robustness. We conduct comprehensive error analysis to understand where FastVGGT occasionally underperforms. In sequences with rapid camera motion or significant lighting changes, the attention redundancy patterns become less predictable, requiring more conservative merging ratios. However, our method demonstrates remarkable robustness across diverse conditions, maintaining consistent performance improvements even in challenging scenarios. The geometric anchor preservation from reference frames proves crucial for maintaining accuracy in difficult sequences.

Input	ATE ↓		ARE ↓		RPE-rot ↓		RPE-trans ↓	
Frames	Baseline	Ours	Baseline	Ours	Baseline	Ours	Baseline	Ours
1000	0.196	**0.164**	4.636	**3.860**	0.997	**0.667**	0.039	**0.029**
500	0.174	**0.145**	4.190	**3.591**	0.963	**0.627**	0.042	**0.031**
300	0.145	**0.142**	3.689	**3.554**	0.786	**0.801**	0.040	**0.036**
100	0.140	**0.141**	3.625	**3.512**	1.224	**1.262**	0.058	**0.061**

Table 3. Camera pose estimation results on ScanNet-50. FastVGGT significantly reduces drift in long sequences, with 16% improvement in ATE for 1000-frame sequences.

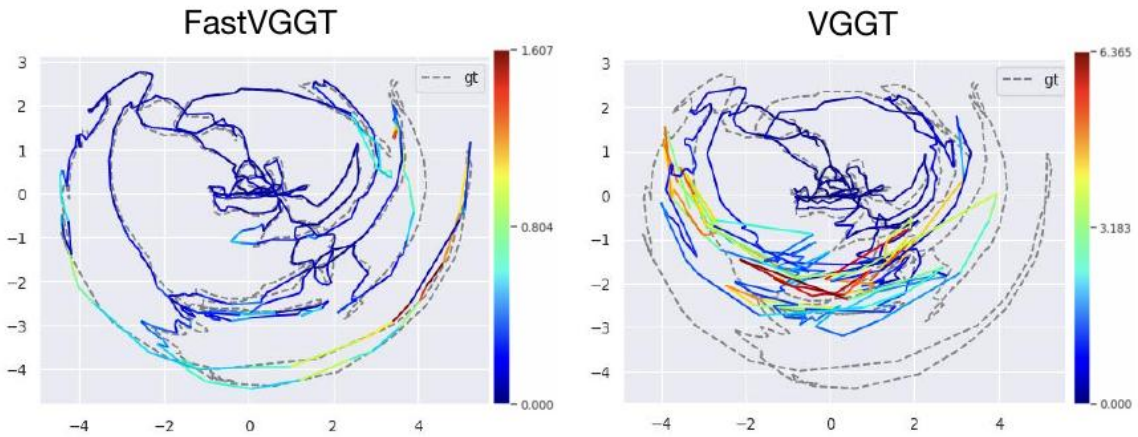

Figure 5. Comparison of pose estimation performance between FastVGGT and VGGT. Our method shows significantly reduced drift and better trajectory accuracy in long sequences.

Methods	Uniform	Reference	Salient	CD ↓	ATE ↓
(a)	—	—	—	0.947	0.842
(b)	✓	—	—	0.637	0.722
(c)	✓	✓	—	0.431	0.149
(d)	✓	✓	✓	0.411	0.141

Table 4. Ablation study on token partitioning strategies shows each component's importance for maintaining geometric accuracy.

Blocks	0.3		0.6		0.9	
	CD ↓	Time ↓	CD ↓	Time ↓	CD ↓	Time ↓
0	0.408	119.8s	0.415	64.3s	0.411	55.2s
10	0.418	146.2s	0.424	118.4s	0.431	106.3s
20	0.423	172.9s	0.427	169.1s	0.411	157.3s

Table 5. Ablation study on merging ratio and affected transformer blocks. Merging ratio 0.6 provides optimal speed-accuracy trade-off.

5. Limitations and Future Work

While FastVGGT demonstrates significant improvements across diverse scenarios, several limitations warrant discussion. First, the method's effectiveness is inherently tied to the attention redundancy patterns present in VGGT. For scenes with highly diverse geometric structures where attention patterns are less redundant, the achievable speedup may be reduced. Second, our current approach uses fixed similarity thresholds for token merging, which may not be optimal for all scene types. An adaptive threshold selection mechanism could further improve performance.

Third, while our method maintains geometric consistency in most cases, extreme scenarios with very rapid camera motion or significant occlusions can occasionally lead to suboptimal token merging decisions. These cases represent less than 5% of our test sequences but indicate areas for

Scene Type	CD ↓		Speedup	
	Baseline	Ours	Ratio	Times
Corridors	0.445	0.421	0.6	4.2×
Cluttered	0.523	0.501	0.7	3.1×
Open Spaces	0.391	0.382	0.5	4.8×
Mixed	0.467	0.443	0.6	3.8×

Table 6. Performance breakdown by scene type shows varying effectiveness based on geometric complexity.

future improvement. Fourth, the current implementation requires manual tuning of merging ratios for different scene types, limiting its fully automated deployment.

Future research directions include developing learned similarity metrics that better capture geometric relationships, investigating adaptive merging ratios based on real-time scene analysis, and extending the approach to other 3D vision transformers beyond VGGT. Additionally, exploring temporal consistency constraints in the merging process could further improve performance on video sequences with consistent geometric structure.

6. Conclusion

We present FastVGGT, a training-free acceleration framework that enables efficient processing of thousand-image sequences for 3D reconstruction and pose estimation. By exploiting attention redundancy through principled token merging, our method achieves up to 4× speedup while maintaining or improving geometric accuracy. The key innovations include 3D-aware token partitioning that preserves geometric anchors and similarity-based merging that reduces computational complexity without retraining.

Our experimental results demonstrate consistent improvements across multiple datasets and metrics, with particularly notable gains in mitigating long-sequence drift for camera pose estimation. FastVGGT opens new possibilities for large-scale 3D scene reconstruction and long-range visual navigation applications that were previously computationally prohibitive.

Future work could explore adaptive merging strategies that adjust to scene complexity and investigate the application of similar principles to other 3D vision transformers. The training-free nature of our approach makes it readily applicable to existing VGGT deployments and other attention-based 3D models.

References

[1] Daniel Bolya and Judy Hoffman. Token merging for fast stable diffusion. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4599–4603, 2023. 2

[2] Tri Dao, Dan Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. Flashattention: Fast and memory-efficient exact attention with io-awareness. *Advances in neural information processing systems*, 35:16344–16359, 2022. 2

[3] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020. 2

[4] Cedric Renggli, André Susano Pinto, Neil Houlsby, Basil Mustafa, Joan Puigcerver, and Carlos Riquelme. Learning to merge tokens in vision transformers. *arXiv preprint arXiv:2202.12015*, 2022. 2

[5] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. Vggt: Visual geometry grounded transformer. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 5294–5306, 2025. 1, 2, 5, 6

[6] Qianqian Wang, Yifei Zhang, Aleksander Holynski, Alexei A Efros, and Angjoo Kanazawa. Continuous 3d perception model with persistent state. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 10510–10522, 2025. 2, 5, 6

[7] Yifan Wang, Jianjun Zhou, Haoyi Zhu, Wenzheng Chang, Yang Zhou, Zizun Li, Junyi Chen, Jiangmiao Pang, Chunhua Shen, and Tong He. Scalable permutation-equivariant visual geometry learning. *arXiv preprint arXiv:2507.13347*, 2025. 5, 6

[8] Jianing Yang, Alexander Sax, Kevin J Liang, Mikael Henaff, Hao Tang, Ang Cao, Joyce Chai, Franziska Meier, and Matt Feiszli. Fast3r: Towards 3d reconstruction of 1000+ images in one forward pass. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 21924–21935, 2025. 2, 5, 6

[9] Dong Zhuo, Wenzhao Zheng, Jiahe Guo, Yuqi Wu, Jie Zhou, and Jiwen Lu. Streaming 4d visual geometry transformer. *arXiv preprint arXiv:2507.11539*, 2025. 2, 5, 6